

Featured Article

Are You Estimating the Right Thing? An Editor Reflects

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Abstract *A key aspect of applied economics over the last decade has been the status accorded to research designs that may credibly identify causal effects. Without such designs, one may not estimate the right thing and publishing options may be limited. Yet situations where reported results are not for parameters that researchers think they have identified are potentially more widespread than what is usually considered in discussion of research designs or identification strategies. I give five examples of applied practice where what is estimated may not identify what is wanted. Each practice came up in submissions I handled as co-editor of an agricultural economics journal, and these issues seem common in applied work. A narrow focus on identification strategies may lead to the neglect of these examples, and of other productive ways to improve the credibility and usefulness of applied economics research.*

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A key aspect of applied economics over the last decade is the status accorded to research designs that may credibly identify causal effects. These designs often focus on estimating treatment effects from randomly assigned treatments, as in a randomized control trial (RCT), or use quasi-experiments, where something correlated with treatment has random variation. RCTs are often regarded as being at the top of a claimed hierarchy, as Imbens (2010) states, “Randomized experiments do occupy a special place in the hierarchy of evidence, namely at the very top.”

While there are prominent critics of this view (e.g., [Deaton and Cartwright 2018](#)), it has influenced recent research design choices made by applied economists. A step below RCTs in the hierarchy would be instrumental variables (IV) and regression discontinuity (RD); both can deal with selection into treatment based on unobservables, and IV helps analyze RCTs with imperfect compliance. Further down would be difference-in-differences (DD) and panel data research designs that rely on parallel trends and time-invariant fixed effects, while lower still, OLS and matching estimators used with observational data are increasingly considered less credible.

Privileging randomized research designs matters to many applied economists because it may affect options for publishing empirical research. As [Ravallion \(2018\)](#) notes, “Researchers who did not use randomization started to be criticized by the randomistas. Some of this took the form of referees’ comments on journal articles. . .”.

In response, a generation of graduate students may emerge who spend much of their doctoral years looking for agencies willing to have clients experimented upon, or searching for policy discontinuities that yield some quasi-randomization. A skew in studying what is amenable to randomization rather than studying topics that are the most important is a common complaint; for example, the structural econometrician, John [Rust \(2010\)](#) writes: “the question doesn’t even need to be important as long as one has a clever instrument . . .”.

My criticism is subtly different. Not only should applied economists study the most important topics, they also should focus on the most binding constraints on our understanding of these topics. Weak research designs may be the biggest constraint, but that has hardly been demonstrated. Very few studies show the size of the bias from using ostensibly worse versus better research designs.¹ Perhaps other problems, like measurement error and misinterpreting data, do more damage to our understanding. If so, a narrow focus on issues of research design may lead to neglect of other productive ways to improve the credibility and usefulness of applied economics research. As economists, we should allocate our efforts to where marginal benefits most exceed marginal costs. My examples below, of applied economists estimating the wrong thing, suggest that we do not practice what we preach. Fixing the issues in these examples may be easier and more productive than switching research designs to higher status ones, like RCTs. In other words, an overly narrow focus on identification strategies may reflect a failure to think like an economist, in terms of optimally allocating attention.

Two broad examples of constraints on understanding will suffice before I give five specific examples of areas where economists seem to estimate the wrong thing. In terms of measurement error, technological advances may make some data more reliable (e.g., Global Positioning System data for measuring plot sizes), but on the other hand, survey measurement of living standards becomes harder as countries rapidly urbanize and people get richer ([Gibson 2016](#)). Yet “measurement error” has gone from being

¹One example that is expanded on below suggests about a 30% bias from using OLS with observational data rather than a research design with randomization. Even so, we might prefer a biased observational study over an RCT if it has sufficiently lower variance under a mean-squared-error decision criteria ([Ravallion 2018](#)).

mentioned eight times more often than “identification strategy” in economics papers 20 years ago, to now having fewer mentions.² Indeed, mentions of measurement error declined 15% over the last decade, even though the number of economics papers grew by more than one-quarter. It is unclear that this reduced attention to measurement error reflects a clear-eyed assessment that this problem is no longer a major constraint on our understanding of important problems in applied economics.

In terms of misinterpreting data, rather than measurement error per se, it is sobering to note that a systematic review of use of one type of data (sub-national data for China), found over 80% of applied studies in three groups of leading economics journals misinterpreted the data (Gibson and Li 2017). Some of these studies had careful identification strategies, with plausibly exogenous variation in covariates, but researchers studied outcomes with data that could not possibly measure what they want, because their data were not about who actually lives or works in a given location, and instead were for who had registered in that location, with a gap of about 300 million people between the two concepts. A little more attention to what data actually measure, rather than fixating on research designs, may have helped these studies.

In the next section, I briefly review what is meant by “identification strategy” and “research design”, and also briefly summarize some existing critiques of privileging the randomized designs. In the following section, I provide five examples of applied practice where what is estimated is unlikely to be what is wanted. None of these five examples are typically thought of when research designs are discussed. Thus, my examples help to make the point that an overly narrow focus on research designs may lead to the neglect of other, potentially more productive ways to improve the credibility and usefulness of applied economics research. Each example came up in submissions I handled as co-editor of an agricultural economics journal, the *Australian Journal of Agricultural and Resource Economics* (AJARE). However, these issues seem common in applied work, and some also featured in papers from academics in business and economics departments that I reviewed as a panelist for a national research assessment. So these examples of estimating the wrong thing are likely to occur quite widely.

The “Credibility Revolution”

Most readers will be well aware of the heightened focus on identification strategies and of the approach of privileging randomized research designs, which is given various names by proponents, including “reduced form modeling”, “causal modeling”, and “treatment effects modeling”. I will refer to proponents of this approach as *randomistas*, a term used by both fans (e.g., Leigh 2018) and critics (e.g., Ravallion 2018). A key claim of *randomistas* is that “[E]mpirical microeconomics has experienced a credibility revolution . . . [whose] . . . primary engine driving improvement has been a focus on the quality of empirical research designs,” (Angrist and Pischke 2010).

²Based on an EconLit search of titles and abstracts of journal articles and working papers, on January 9, 2019, for three three-year periods of 1996–98, 2006–08, and 2016–18, with “measurement error” mentioned, respectively, in 199, 552, and 469 papers, while “identification strategy” was mentioned, respectively, in 26, 204, and 514. Perhaps future research will return to measurement issues; for example, measurement and survey design are given prominent attention in the Development Impact Blog, which provides a lot of resources for researchers who use randomized designs.

There is no single, seminal source for these developments but they are generally agreed to have emerged first in labor economics, and then influenced practice in other areas of applied microeconomics. A good source document, therefore, is the widely cited chapter on empirical research strategies in the *Handbook of Labor Economics* (Angrist and Krueger 1999), which outlines an “experimentalist approach” that “... puts front and center the problem of identifying the causal effects from specific events or situations ... Here, the ideal research design is explicitly taken to be a randomized trial and the observational study is offered as an attempt to approximate the force of evidence generated by an actual experiment.”

Related to “research design” is the “identification strategy” (Angrist and Pischke 2010 use the two terms interchangeably). While *identification* was used in econometrics before (Lewbel 2019), it was given a new meaning by proponents of the experimentalist approach. The “identification strategy” is (Angrist and Krueger 1999) “... the combination of a clearly labeled source of identifying variation in a causal variable and the use of a particular econometric technique to exploit this information ...”.

The focus on identifying variation in covariates that might enable one to claim that “ x causes y ” reflects concerns with previously popular empirical approaches such as OLS regressions with observational data. To measure the impact of something (a “treatment”) we would ideally see the same individual with and without the treatment. This is impossible, given that we have just one existence, so there is a problem of missing counterfactual information. An issue for observational studies is that people whose potential outcomes in the treated state exceed their potential outcomes in the untreated state will tend to self-select into the treatment. Naïve estimates of treatment effects, such as an OLS regression of outcomes on a dummy variable for people in the (self-selected) treatment group, will mistake pre-existing differences in potential outcomes for causal effects of the treatment. With randomization, the average outcomes for the untreated (control) group should be a credible guide to the average of the potential outcomes in the (counterfactual) untreated state for the people who actually were treated, and so the missing information need not cause bias.

An illustrative example is the impact of migrating from a poor country to a rich one, which is a highly selective treatment (just 3% of people live outside their birth country). Causal effect estimates that use the randomization provided by a migration lottery suggest that work incomes rose 263% within a year of making such a move (McKenzie et al. 2010). If people self-select into migration, based on unobservable (to researchers) characteristics that the labor market values, designs without randomization tend to overstate migration impacts by also including returns to these unobservables. Thus, when McKenzie et al. (2010) use OLS with observational data from the same setting, the apparent migration impact is overstated by about 30%; using various matching techniques, the overstatement is by about 25%; and, with difference-in-differences the overstatement of impacts is by about 21%. If instrumental variables are used, with a plausible instrument that should satisfy exclusion restrictions, the overstatement is only by 13%, and by as little as 1% in a sub-sample.³ Thus, the less credible research designs tend to

³If an instrument that is unlikely to satisfy exclusion restrictions is used, the migration impact estimated using IV is overstated by 82%, emphasizing the importance of having plausible instruments.

overstate estimated impacts of the treatment the most, and this is one sense in which they may not estimate the right thing.

Yet this example is quite atypical because very few studies show the size of the bias from using ostensibly worse versus better research designs. Given the potential cost of the randomized designs, including in ethical terms, it would be helpful if the magnitudes of both their benefits and costs were more clearly set out by their proponents. Other criticisms of the *randomistas* are as follows: (a) that attention focuses on less ambitious sets of questions such as treatment effects of a particular policy in place, but not effects of new policies or of old policies in new environments, creating a selection bias in what gets evaluated; (b) that randomized designs make no fewer assumptions than structural econometrics, but do so implicitly not explicitly, and with assumptions about determinants of treatment, such as “no defiers” (subjects doing the opposite of what they are assigned to do) rather than about how treatment affects outcomes; (c) that *randomistas* focus on *whether* treatments worked, but not *why* they worked; (d) that actual balance in any single RCT is confused with balance in expectation over many (hypothetical) trials; and, (e) that cumulative scientific progress is undermined by minimizing the role of prior knowledge. Key critics with an econometric emphasis include Heckman (2008), Keane (2010), Rust (2010), Deaton (2010), Deaton and Cartwright (2018), and Lewbel (2019), while critics with more focus on agricultural and development economics are Barrett and Carter (2010) and Ravallion (2018).

Some criticism is specific to Instrumental Variables (IV)—the most widely used econometric technique in the *randomista* toolkit—because it is needed both for RCTs with imperfect compliance (as most RCTs have) and also for various quasi-experiments. Amongst these criticisms are the following: (a) that IV relies on a monotonicity assumption, where the instruments shift subjects in only one direction (Keane 2010); (b) that the local average treatment effect (LATE), which is the effect of the treatment on compliers, will typically not be either the average treatment effect or the average treatment effect on the treated and it is *ex ante* unclear who are members of the set over which the LATE averaging is done (Deaton 2010); and, (c) that if different instruments yield different parameter values, then under structural identifying assumptions at least one instrument cannot be valid, but under the causal reduced form interpretation placed on IV by the *randomistas*, different instruments yield different parameter values because they correspond to different estimands, each considered to be causally meaningful because each equals the (local) average treatment effect for some (albeit unknown) set of people in the population (Lewbel 2019).

A Smörgåsbord of Examples

Using Latent Variable, Binary Outcome, Models with Continuous Data

An old issue, first discussed by Ravallion (1996) but still showing up in applied work, is researchers estimating latent variable, binary outcome models, such as probits, when they have continuous and fully observed data. For example, they may have data on household consumption but they estimate a probit model of the probability that households are below the poverty line. Specifically, for per capita consumption c , poverty line z , and covariate vector x , they model $\text{prob}[c < z \mid x]$. Such modeling involves

ignoring information on the continuous distribution of consumption, pretending one knows only a binary variable for whether households are poor or not. The motivation for this may be a desire to see how covariates change the risk of a particular outcome (e.g., the risk of being poor), once marginal effects are calculated from the probits.

This procedure is wrong for at least three reasons. First, latent variable models make distributional assumptions about the unknown errors that are not needed for models like OLS (and Instrumental Variables) that can be estimated on the continuous data. The consequences of misspecification are worse with binary outcome models; for example, heteroscedasticity makes maximum likelihood estimates of probit coefficients inconsistent, while OLS stays consistent, with heteroscedasticity only affecting efficiency of OLS.

Second, using latent variable models when one wants to interpret results in terms of how covariates change the probability of an outcome is redundant, as these probabilities can be derived from OLS (and IV) results. For example, the risk of being poor can be estimated after an OLS regression, for logged consumption over the poverty line, using

$$\text{prob}(\ln(\hat{c}_i/z)) < 0 = \Phi\left(\left(-x_i\hat{\beta}\right)/\hat{\sigma}\right).$$

The *Stata* command: `predict poor, pr(., 0)` issued after OLS (or IV) creates a variable ("poor") with the estimated probability that (log) real consumption is in the interval from negative infinity to the poverty line for each household. No distributional assumptions need be made for the regression estimators that provide the values used to produce these probability estimates, unlike for probit or logit models.⁴ These estimating equations also are robust to misspecification that can make binary variable models inconsistent.

Third, probits are biased by random errors in dependent variables, unlike for linear regression (e.g., OLS) where dependent variable errors only cause imprecision but not bias (Hausman 2001). Random errors cause misclassification in binary variable models, where a dependent variable has a value of one (e.g., a household is defined as poor) when it should have the value zero, or vice versa, and such errors can lead to large biases. For example, Hausman et al. (1998) show that with just a 5% risk of misclassification, the probit coefficient on a right-hand side (RHS) dummy variable (e.g., for a female-headed household) falls to 70% of its true value. It is even worse for a log-normally distributed RHS covariate (e.g., land holdings); with just a 2% (5%) risk of misclassification, the coefficient on this variable is biased down by more than 20% (45%).

The redundant and probably biased approach of using latent variable, binary outcome models when a researcher has continuous data should be distinguished from legitimate debate about whether OLS (strictly, the Linear Probability Model) or non-linear models like probits and logits are better when one has actual binary variables (e.g., whether a plant closed down

⁴Complex sample designs are considered below, such as the clustered samples widely used for household surveys. The correct estimators for such data, such as the *svy* commands in *Stata*, may not allow these probabilities to be estimated after a survey regression estimate because $\hat{\sigma}$ may not be constant in clustered samples. A workaround is available, using pseudo-maximum likelihood, for example, with the survey interval regression estimator (which also allows point data). See Gibson and Rozelle (2003) for details.

or not).⁵ The school of thought associated with *Mostly Harmless Econometrics* (Angrist and Pischke 2009) argues for “just use OLS” because if the conditional expectation function is linear then a linear model like OLS (or IV) will provide this, and if it is non-linear then a linear model usually approximates it, especially if focus is on the marginal effects rather than on the coefficients for the latent index variable. Also, there are advantages with linear models for inference and in not having to choose if marginal effects are measured as finite differences or as derivatives. Abandoning the practice of using latent variable, binary outcome models when one has continuous data would be an easy way to improve the credibility of results that are framed in terms of the probability of certain outcomes, such as being poor.

Confusing Quantity Response to Price with the Quality Response to Price

Consumers usually reduce quantity consumed if prices rise. Estimating the elasticity of this response is a staple for agricultural economists. Consumers also may downgrade the quality of what they consume; for example, if beef prices rise, a switch from filet mignon to cheaper cuts is likely. Even a researcher only interested in the response of quantity to price needs to deal with the quality response in case it contaminates their estimates. This concern with correctly identifying a quantity demand elasticity especially matters to research using household survey data.⁶ It has long been recognized that data from these surveys mix together items of differing quality. According to (Prais and Houthakker 1955), “An item of expenditure in a family-budget schedule is to be regarded as the sum of a number of varieties of the commodity each of different quality and sold at a different price.”

To show this quality variation, for a food group that receives a lot of policy and research attention, I surveyed all the carbonated (fizzy) soft drinks in my local supermarket. The 160 specifications on the shelves had a 14:1 price range, from \$9.20 per liter (a 250 ml can of *Red Bull*) to \$0.66 per liter (a 1.5 liter bottle of store-brand cola). Just for colas there was a 9:1 price range. This huge variation gives scope for a consumer to maintain the quantity of their fizzy drink intake if prices were to rise, by sliding down the quality scale from expensive *Red Bull* (or even expensive cola) in small cans to cheap store-brand cola in large bottles.

Many studies that use household survey data to ostensibly estimate price elasticities of quantity demand ignore this type of within-group quality variation. A systematic review suggests that over 80% of published demand studies using household survey data mix together consumer responses on the quantity margin with responses on the quality margin, thus overstating the rate that quantity demand responds to price (Gibson and Kim 2019a). In part, this is because many applied economists yield to the temptation to divide survey estimates of household expenditures on a food group by the reported quantity bought for that group, and call the ratio a “price” and use it to estimate elasticities. Phrases such as the following indicate such procedures: “prices have been calculated by dividing expenditure over quantities . . .”

⁵A good discussion of this debate is covered here: <https://blogs.worldbank.org/impacetevaluations/whether-to-probit-or-to-probe-it-in-defense-of-the-linear-probability-model>.

⁶It also matters to scanner data if transactions for multiple Universal Product Code (bar-code) items are collapsed into broader categories like “sports and energy drinks” (as done, e.g., by Zhen et al. 2011).

Such studies are wrong for two reasons, one for using unit values (expenditures over quantities), and one that happens even if researchers use actual prices. First, unit values poorly measure prices over time and space due to the Alchian-Allen effect (or “shipping the good apples out”), which causes the relative price of quality to vary over time and space due to shipping and storage costs. For example, if *Red Bull* and store-brand cola were shipped 100 miles from my local supermarket, the relative price ratio would then be less than 14:1 due to transport costs. This matters as countries commercialize their agriculture because foods then travel further through time, space, and form (e.g., processing) than when most people ate self-produced foods. If the relative price of quality within a survey group changes over time and space, the composition of demand within that group should also change, so unit values in one period or place will not refer to the same quality mix as unit values elsewhere. Consequently, unit values cannot be a consistent indicator of the group price level over time and space. Evidence for these effects, and the distortion caused when unit values are wrongly used as a proxy for prices, is reported by [Gibson and Kim \(2015\)](#) and [Gibson \(2016\)](#).

Second, even if market prices (or adjusted unit values meant to mimic them, as in [Cox and Wohlgenant 1986](#)) are independent variables, if the dependent variable is budget shares, such as for the fizzy drinks group, those shares vary with the consumer’s choice of quantity and quality ([McKelvey 2011](#)). Thus, typical budget share models, like the Almost Ideal Demand System or the Linear-Approximate (LA-AIDS), will identify a hybrid of the price elasticities of quantity and of quality. In these studies, w_{Gi} , the share of the budget devoted to food group G by household i , is modeled as varying with log per capita expenditures, $\ln x_i$, with the log of the price index for foods in group H (with θ_{GH} for the cross-price effect and θ_{GG} for the own-price effect), with conditioning variables, z_i , and with random noise, u_i .⁷

$$w_{Gi} = \alpha_G^0 + \beta_G^0 \ln x_i + \sum_{H=1}^N \theta_{GH} \ln p_H + \gamma_G^0 \cdot z_i + u_{Gi}^0.$$

The typical study then bases the reported elasticity of quantity with respect to price on:

$$\varepsilon_{GH} = (\theta_{GH}/w_G) - \delta_{GH},$$

where δ_{GH} equals 1 if $G=H$, and 0 otherwise. This is correct only for standardized items with no quality variation possible, or for aggregate goods (e.g., a time series study) with the same quality composition over time. For those, the numerator of the budget share is the product of price and quantity for good G , so if the budget share is differentiated with respect to price, the only possible thing that can be identified is the response of quantity to price.

But as the 14:1 price range for fizzy soft drinks in my local supermarket shows, food groups of interest have large quality variation. If fizzy drink budget shares are differentiated with respect to price, one cannot know whether a quantity response, a quality response, or some hybrid of the two

⁷Superscripts are used to distinguish the same parameters in a soon-to-be-introduced second equation for unit values, which is to allow for the consumer’s quality response to price. Most studies ignore this second equation.

is produced. The solution is to have an equation for each type of response, to identify a price elasticity of quantity demand free of any effects of consumer responses on the quality margin. This two equation framework is due to Deaton (1990), with McKelvey (2011) being the first empirical study using this approach. The second equation is

$$\ln v_{Gi} = \alpha_G^1 + \beta_G^1 \ln x_i + \sum_{H=1}^N \psi_{GH} \ln p_H + \gamma_G^1 \cdot z_i + u_{Gi}^1$$

with the log unit value as the dependent variable. Conditional on prices, the unit value shows where on the quality scale the consumer buys. Log-differentiating with respect to price gives

$$\partial \ln w_G / \partial \ln p_H = \theta_{GH} / w_G = \varepsilon_{GH} + \psi_{GH}$$

with ε_{GH} being the elasticity of quantity demand with respect to the price of H (the parameter of interest), and ψ_{GH} being the elasticity of the unit value with respect to the price of H . In other words, budget shares depend on both quantity effects (ε_{GH}) and quality effects (ψ_{GH}). Rearranging this equation shows why studies typically do not identify the elasticity of quantity demand with respect to price, even though researchers say they want this parameter:

$$\varepsilon_{GH} = (\theta_{GH} / w_G) - \psi_{GH}.$$

The typical approach, with just budget share equations estimated on household survey data, cannot estimate ψ_{GH} and so arbitrarily assumes that this parameter equals 1 for own-prices (i.e., unit values move one-for-one with prices) and 0 for cross-prices. That is why the second equation (for unit values) is needed, so that ψ_{GH} can be estimated not assumed.

This is known in development economics but seemingly not agricultural economics. The first evidence on bias in the typical approach found quantity demand elasticities were inflated an average of 250% of their value, if quality response to price is ignored (McKelvey 2011). That evidence was for just six food groups, somewhat broadly defined (e.g., “meat and fish”, “vegetables”) and only from a small area in Indonesia, so it may not convince everyone. Corroboration comes from a recent study, with prices and qualities observed over all of Vietnam, that found a four-fold exaggeration in price elasticities of quantity demand if quality responses are ignored, irrespective of whether analysis was based on ten broad consumption groups or 45 narrower items within those groups (Gibson and Kim 2019a).

That responses on the quality margin mediate effects of price on quantity matters to sin tax policies that aim to cut consumption, and to policies that compensate for price shocks. For example, Gibson and Kim (2019b) estimate price elasticities of quantity, and of quality, for tobacco products in Papua New Guinea. Higher excise taxes reduce quantity demand by just one-third of what is predicted if quality response to price is ignored. For the second effect, Gibson and Kim (2013) show that periodic rice export bans by Vietnam, which aim to lower local rice prices so that consumers can afford enough calories, may be unnecessary. Price elasticities of quantity demand from

typical methods, which ignore quality response to price changes, give an elasticity of calories with respect to rice prices of -0.5 . With such an elasticity, policy makers might be concerned that high rice export revenue is at the cost of calorie shortfalls for local consumers. But once it is recognized that consumers downgrade quality to cope with price rises, the elasticity of calories with respect to rice prices is not even -0.2 , implying much less of a trade-off between export revenue and consumer nutrition.

A topical example where quality responses undermine results of applied economic policy analyses is Mexico's peso-per-liter tax on soda. As prices rose from the tax, consumers downgraded quality. This type of response is ignored in existing analyses. The quality effect is seen in the figure below, for percentage changes in soda prices and unit values. The unit values are from Mexico's biennial household expenditure survey (ENIGH), based on the surveys immediately before and after the tax. The prices are from city-average monthly prices that Mexico's statistics institute (INEGI) gathers in 46 cities.⁸

Across all the cities, average soda prices rose 11.9% (Figure 1); like other estimates of 12% to 14% post-tax increases (Grogger 2017). Unit values rose only half as much, by 6.3%, on average. This is consistent with consumers downgrading quality to cope with higher prices—by switching brand (e.g., *Pepsi* is about 15% cheaper than *Coke*), by buying larger bottles that are cheaper per liter, and by other shopping strategies (see Griffith et al. 2009 for more on these). Yet most studies assume that unit values move one-for-one with prices ($\psi_{GG}=1$), ignoring responses on the quality margin. With budget share models, a smaller spending increase due to quality downgrading gets wrongly treated as a cut in quantity, unless the two-equation framework of Deaton (1990) and McKelvey (2011) is used. Alternatively, the change in average quantity (which declined 4.6% from 2012 to 2014) may be divided by change in average price to give an arc elasticity. Given that unit values rise less than prices (due to quality downgrading), they give an exaggerated own-price elasticity of -0.85 . But with the actual (larger) rise in city average prices as the denominator, the elasticity is not even half as large, at -0.39 .

The figure also shows movements in prices and unit values for the cheapest, middle, and most expensive tercile of cities (based on long-run soda prices). The soda tax is specific not ad valorem, so percentage price rises are highest in cheaper cities, and the ratio of unit value increase to price increase is the lowest there. This is consistent with consumers who face bigger relative price rises making bigger quality adjustments. There is also a spatial, long-run response to price differences; in the most expensive tercile of cities, average unit values are only 77% of average prices, while in the cheapest cities they are 95% of average prices. Thus, where prices are highest, consumers have, on average, downgraded quality more. These temporal and spatial effects undermine studies using unit values to proxy for prices and undermine studies that ignore consumer responses on the quality margin.

The elasticity bias flows through to policy analyses. For example, Grogger (2017) used prior own-price elasticities of quantity demand for soda in Mexico, to turn tax-induced price rises into bodyweight effects (for health outcomes); with elasticities of -1.0 and -1.3 , Grogger forecast that the soda tax would reduce steady-state body-mass index (BMI) of Mexicans up to

⁸The unit values are only for ENIGH households in cities with the price survey. The average prices are just for the same periods that the ENIGH survey is fielded. I thank Mabel Andalon for these estimates.

Figure 1 Percentage changes in city average soda prices and unit values from Aug-Nov 2012 (pre-tax) to Aug-Nov 2014 (post-tax)



1.8%. If those elasticities were correct, the post-tax soda quantity should have fallen 12% to 15%, all else being equal, yet various data show a fall of just 4% to 7% (e.g., 4.6% using ENIGH). Demand fell much less because those elasticities wrongly mixed together quality responses to price and quantity responses to price. If the correct two-equation framework of Deaton and McKelvey is used, the own-price elasticity of quantity demand for soda is just -0.3 (Andalón and Gibson 2017). With this four-fold elasticity bias from ignoring quality responses to price it is possible that the health benefits of the soda tax will be just one-fourth of what is forecast.

Zeros, Tobits, and Unconditional Expected Values

Reports of zero spending on a food group during the reference period of a household survey are common in applied demand analyses. For example, the ENIGH survey in Mexico referred to above had 34% of households report zero purchases of soda in the week they were surveyed.⁹ Micro data on other topics, like adopting new technologies, participating in certain marketing channels, or migrant remitting also often have lots of zeros. In these cases, some researchers seem to follow the mantra “zeros mean OLS is biased, must do Tobit.”¹⁰ Whether this is right depends on the aim of the study and many authors are vague about their purposes and so it is not clear they are estimating the right thing.

To fix ideas, recall that the Tobit model is

⁹Soda is storable, as are many other foods, so zero purchases in a survey period does not mean consumption is zero. In a survey measuring starting and ending food stocks, 8.3% of household-food group combinations had zero spending and consumed out of stocks during the survey; for some foods this proportion was as high as 17% (Gibson and Kim 2011). If consumers buy when prices temporarily fall and stockpile to consume later, Wang (2015) shows that typical own-price elasticities of quantity demand (for soda) that ignore this effect are exaggerated by up to 60%.

¹⁰This also includes the generalization of Tobit, the sample selection model that lets coefficients for the selection and regression parts of the model differ.

$$y_i^* = X_i\beta + \varepsilon_i \quad \varepsilon \sim N(0, \sigma^2)$$

where y_i^* is a latent variable, such as the desired spending on soda, and this leads to a realized purchase only if the latent index variable exceeds zero, so the observed data are

$$y_i = \begin{cases} y^* & \text{if } y^* > 0 \\ 0 & \text{if } y^* \leq 0 \end{cases}$$

Three different regression functions are associated with the Tobit model, and therefore there are three different types of marginal effects:

- a. The effect of X on the latent index. These are the Tobit coefficients reported by statistical software. These may be of interest to someone wanting to manipulate latent desires, like an advertising executive. But as [Angrist and Pischke \(2009\)](#) note, in calling the latent index a “puzzling construct”, expenditure really is zero for some people, and this is neither a statistical artefact nor due to some kind of censoring. So it is unclear that Tobit coefficients are what is wanted in many studies, even though they are what is reported.
- b. The effect of X on the observed y , conditional on being a non-limit observation. An example would be a regression whose covariates are determinants of spending on soda (not latent spending), estimated just on the 66% of households in the ENIGH sample that reported buying soda in the week they were surveyed. This regression needs an extra term—the inverse Mills ratio—to account for possible selection bias, in the sense that purchasers may be a non-random subset of the population.
- c. The unconditional expected value of y as X changes, which takes into account the probability of being above the limit, and the expected value conditional on being above the limit. To get this effect, the Tobit coefficients need to be multiplied by the fraction of non-limit observations in the sample ([McDonald and Moffitt 1980](#)). For example, if the Tobit coefficients in (a) are multiplied by 0.66, for the ENIGH data, this adjustment would give an effect of X on the unconditional expected value of soda expenditure, averaging over purchasers and non-purchasers alike.

For many studies, the marginal effects from (c) are the right ones and the Tobit ones are not. Continuing the Mexico example, if the Ministry of Finance wants to forecast how much revenue is raised from the soda tax, marginal effect (c) is the relevant one—a peso is a peso is a peso—it does not matter if changes in revenue from the soda tax come from consumer adjustment on the intensive margin or the extensive margin. Similarly, if research is taking results from a Tobit model (or a more generalized sample selection or censored demand model) and feeding them into some sort of population health model, as might be done in an economic-epidemiological study, the marginal effects from (c) should be used.

Rather than multiplying Tobit coefficients by the fraction of non-limit observations in the sample, just running OLS on all observations (the zeros and the non-zeros) gives a very good approximation ([Greene 1981](#)). Thus, the kneejerk reaction to use Tobit if there are a lot of zeros may be wrong. How wrong it is depends on the proportion of non-limit observations; if it

is one-half, then reported marginal effects may be twice as big as they should be.

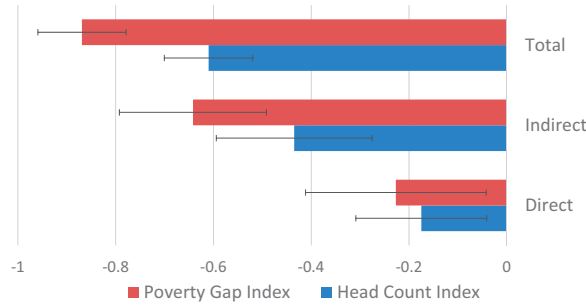
Deaton (1997) provides an extended example that makes this same point: “A good example is the effect of a change in tax rates on tax revenue. A government is considering a reduction in the subsidy on wheat (say), and needs to know the extent to which demand will be reduced at the higher price. The quantity of interest is the effect of price on average demand Some households buy positive quantities of wheat, and some buy none, a situation that would seem to call for a Tobit. Estimation of the model yields an estimate of the response of quantity to price for those who buy wheat. But the policymaker is interested not only in this effect, but also in the loss of demand from those who previously purchased, but who will drop out of the market at the higher price. These effects will have to be modeled separately and added into the calculation. But this is an artificial and unnecessarily elaborate approach to the problem. The policy question is about the effect of price on average demand, averaged over consumers and nonconsumers alike. But this is exactly what we would estimate if we simply regressed quantity on price, with zeros and nonzeros included in the regression.”

Ignoring Spillovers when Studying Inherently Spatial Phenomena

Agricultural economists deal with inherently spatial phenomena; space is a key input in production and environmental heterogeneity matters a lot. Coordination by nearby agents also gives spatial patterns; for example, clearing forest from one area raises the likelihood of neighbors clearing. Spillovers create estimation and interpretation issues that many studies ignore. An estimation issue is “I am my neighbor’s neighbor”, so own-effects and spillover effects need to be estimated simultaneously, different to the usual situation for time lags where the past affects the future but not the reverse. An interpretation issue is that the total effect of a covariate may exceed what is shown by standard methods, like OLS, that treat observations as independent and ignore potential spillovers.

Spillovers can be allowed for with a spatial weights matrix, W , that has a value of one for neighbors (e.g., areas sharing any point on the border with area i), and zero otherwise. The diagonals of W are zero because an area cannot be its own neighbor. The spatial weights matrix allows spatial lags to be created, which are averages of a variable in neighboring areas. The lags may be of the dependent variable, the independent variable(s), or the errors. The type of spatial lags relevant to a given setting indicate whether spillovers operate locally or globally. For example, a change in covariate X may cause a local change in the outcome, which then affects the outcomes of neighbors, in turn, affecting outcomes of their neighbors, including the original area. In addition to global effects operating through the spatial lag of the dependent variable, spatial lags of independent variables let changes in X affect outcomes of local neighbors. With spillover and feedback effects, the impact of a change in a covariate in area i on outcomes in area j may differ for each i - j combination. These various impacts can be decomposed into direct and indirect components, following LeSage and Pace (2009): the *direct effect* is the effect of a change in a covariate in area i on the dependent variable in area i , averaged over all areas; the *total effect* is the effect of the same change in the covariate in all areas on the dependent variable in area i , averaged over all areas; the *indirect effect* is the difference between the total effect and direct effect.

Figure 2 Direct and indirect effects of urban growth on rural poverty in India



Depending on geography, transport networks, other infrastructure, and population patterns, indirect effects might exceed the direct effects. If so (and if the sign is the same), ignoring spillovers could lead to understating the impacts on outcomes of changes in the covariate.

For example, a study of how urban growth (proxied by night lights) cut rural poverty found that indirect effects were almost three-quarters of the total effect (Gibson et al. 2017). This study used a four-wave panel of 59 regions covering all India; a country where poverty rates and poverty reduction are non-random over space.¹¹ Part of the indirect effect came through spatial lags of the rural poverty rate, reflecting that it is hard to escape poverty if neighbors are poor, and conversely, local economic growth is easier if neighbors are rich. The results in Figure 2, for the elasticity of the headcount poverty index and of the poverty gap index (which measures the depth of poverty) with respect to urban growth, show direct effects of only about -0.2 , but indirect effects are from -0.4 to -0.6 . The indirect effects came from urban growth in nearby regions lowering rural poverty in region i , and from spillovers amongst the rural poverty rates. Thus, if these effects were ignored, as would happen with using OLS, for example, only about one-third of the total effect of urban growth on rural poverty would be captured, and an important contribution to better outcomes in rural India—from stronger rural-urban linkages and faster post-reform urban growth—would be missed.

A useful approach would be to start modeling with a general spatial model, which has OLS and other aspatial models as special so that, spillovers can be allowed if they show up as relevant. This general-to-specific modeling strategy has inferential advantages (Hoover and Perez 1999). To illustrate, I assume that one has panel data, so that μ_i time-invariant fixed effects for each area, and ϑ_t year fixed effects can also be included. The spatial autoregressive model with spatial autoregressive errors (SARAR) is:¹²

$$Y_{it} = \lambda WY_{it} + \beta_1 X_{it} + \beta_2 WX_{it} + \mu_i + \vartheta_t + \rho Wu_{it} + e_{it}.$$

¹¹Low rural poverty rates are concentrated in the northwest and along the Arabian sea coast, while the south and east have high rural poverty rates, except for a couple of areas around big cities.

¹²This model can be estimated by quasi-maximum likelihood based on Lee and Yu (2010) using a recently implemented estimator in Stata (Drukker et al. 2013).

This model lets changes in an outcome variable in a particular time and space affect contemporaneous outcomes in other areas (via the spatial lag of the dependent variable, if $\lambda \neq 0$) and changes in independent variables affect not only outcomes in the same area but also in neighboring areas (if $\beta_2 \neq 0$). The SARAR nests the spatial Durbin model if $\rho = 0$, where the dependent variable and independent variables are lagged but the errors are not, the spatial auto-regressive model (also called the spatial lag model) if $\beta_2 = \rho = 0$, where only the dependent variable is spatially lagged, and the spatial error model, where only the errors have a spatial lag (if $\lambda = \beta_2 = 0$).¹³ The SARAR also nests aspatial models, such as the fixed effects panel estimator without any spatial lags (if $\lambda = \beta_2 = \rho = 0$).

The SARAR model should provide unbiased coefficient estimates even if one of the nested models is the true data-generating process. The reverse is not true, given that it involves omitting relevant variables. The major computational burden is not so much the estimation of the SARAR model, but the calculation of the impact effects if one has a large sample size because these impact effects potentially differ for each i - j combination of areas. However, approximation methods are available to speed up these calculations. It should also be noted that while the example given here is spatial, spillovers also occur in other networks, such as social networks. The data available to applied economists are increasingly “network-referenced” (as a more general term than “geo-referenced”) and so using models that can handle this complexity will help to better estimate the total impacts of shocks to covariates.

Ignoring Complex Sample Designs

Many empirical studies in agricultural economics rely on samples with at least some of the following features: weights, where some sample observations represent more members of the population than do others (see [Kish 1992](#) for a discussion of four sources of unequal weights); multi-stage sampling, where Primary Sampling Units (PSUs) are first selected and then elements (depending on the topic of interest, these may be households, persons, firms, farm-level plots, etc) within those PSUs are selected and surveyed; and stratification.

Weights may be either by design, to get larger subsamples for groups of interest, or to restore representativeness in the case of non-response (up-weighting observations from areas or groups with low response rates). Multi-stage samples are used for cost-effective fieldwork; one can more cheaply survey a sample of 100 by first selecting ten villages to visit and then getting ten households in each rather than selecting 100 households at random (which likely entails traveling to far more than ten villages). Explicit stratification may occur if survey designers use prior information on factors associated with the outcome of interest (such as remoteness, for a poverty study) to draw a sample in accordance with population proportions rather than leaving this to chance. These features, generally described as “complex samples”, mean that equal selection probabilities may not hold and observations may not be independent.

¹³The spatial error model does not allow spillovers. Thus, it provides a potential improvement in inferences over OLS, but it does not alter the interpretation of results to the same extent as do the models that allow spillovers.

Yet, many applied papers ignore some or all of these features of their samples when reporting on econometric results.¹⁴ Instead, they typically reported some form of regression estimate (which may or may not be weighted, although the focus in this section is more about variances than about point estimates) with heteroscedasticity-robust standard errors. The origin of these standard errors can be attributed to either Eicker (1967) or especially Huber (1967), but economists tend to cite White (1980) for the following variance estimator:

$$V(\hat{\beta}) = n(X'X)^{-1} \left(\frac{1}{n} \sum_{i=1}^n e_i^2 x_i x_i' \right) (X'X)^{-1}$$

compared to $\sigma^2(X'X)^{-1}$ with homoscedasticity

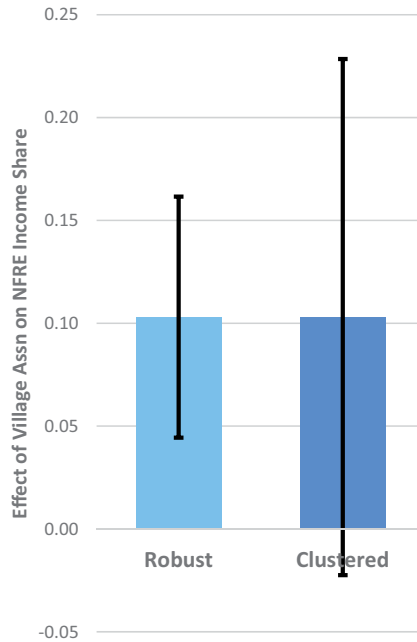
where e_i is the i th least squares residual. A first, minor note is that these standard errors are robust to arbitrary and *unknown* forms of heteroscedasticity, but with a clustered sample we should know something about the form of heteroscedasticity. If observations within a cluster are more alike than are observations from different clusters, this should give a form of grouped heteroscedasticity where σ^2 differs cluster-by-cluster.

The bigger problem is that the variance formula treats observations as independent and so overstates precision by ignoring any correlation between observations sharing the same cluster. Intuitively, there is less variability in a clustered sample with N observations than in a simple random sample (SRS) of the same N , but the robust standard error formula does not recognize this. The lack of independence follows from the multi-stage cluster sampling. For example, a sample may have households from, say, 16 villages, amongst the 1,400 villages in a province like Shanxi in China. These villages are selected during one stage in a multi-stage sample design where, perhaps, four counties were first selected, with a rich and poor township per county then chosen, and some method then used to select two villages per township. With this design, once the 16 villages are selected, households in the other 1,384 villages in Shanxi have zero probability of being in the sample, while once the first household in a village is selected, others in the same village have a non-zero (dependent) probability of being in the sample. Even if many households (say, an average of 30 per village) are surveyed, to yield a sample of almost 500 households, the gain in statistical precision usually obtained from averaging across independent observations does not occur because of correlations between households in the same village.¹⁵ A somewhat conservative standard is that degrees of freedom with clustered samples depend only on the number of clusters (minus the number of strata), so this example has (at most) 16 denominator degrees of freedom. This reflects the idea of no

¹⁴Ignoring these features might be acceptable if one is only interested in learning about the particular sample, and not in making inferences about the population it is drawn from, as noted in the discussion here: <https://blogs.worldbank.org/impactevaluations/when-should-you-cluster-standard-errors-new-wisdom-econometrics-oracle>. However, the motivating statements made in the introductions of most empirical papers imply that most researchers are interested in making inferences about the population.

¹⁵Ignoring weights also understates variances, especially if weights are unequal, because statistics have a higher dependence on a few highly weighted sample points than in an unweighted sample of the same size. Conversely, ignoring stratification will overstate variances. However, the big bias is almost always from ignoring clustering.

Figure 3 Robust standard errors disguise imprecise effects of village associations on income share from non-farm rural enterprises



new independent information from the second and subsequent households surveyed in a village, compared to the new information that would come from the first household surveyed in another village.

An early, cautionary, example of the inference errors that result when clustering is ignored is provided by [Pepper \(2002\)](#), using the Panel Study on Income Dynamics (PSID). The 2930 original PSID families were selected from just 80 counties (out of the more than 3000 counties in the United States), where counties were used as the first-stage PSU so as to reduce interviewing costs. The standard error on the years of schooling term in a log-wage equation is 37% higher when this county-level clustering is accounted for, compared to when the heteroscedasticity-robust standard error is used. A term that should be more widely used in agricultural economics is the *design effect* (DEFF), which is the ratio of the variance estimated when complex sample designs are accounted for, to the variance estimated if SRS is wrongly assumed. In the [Pepper \(2002\)](#) example, the DEFF on the years of schooling term in the wage equation is 1.89, while an inherently geographically clustered variable, location in the South, has a DEFF of 3.2.

Many things of interest to agricultural economists will have higher DEFFs, especially in poor countries where livelihoods are more geographically clustered than they are in the United States. For example, [Gibson et al. \(2014\)](#) use a sample of 1,600 households selected from 97 clusters (*desa* or villages) in Indonesia, to study the household-level and village-level determinants of income shares from non-farm rural enterprises (NFRE). The clustered standard errors are roughly double the size of the heteroscedasticity-robust standard errors, implying DEFFs of about four.¹⁶

¹⁶Analyses at the plot-level, rather than the household level, likely have higher DEFFs. For example, mean plot-level output from the Malawi Third Integrated Household Survey (a World Bank LSMS

The biased inferences that would result from ignoring this clustering create the illusion of apparently statistically significant ($p < 0.05$) effects; for example, Figure 3 shows the effect of having a village business association on the NFRE income share, with various other factors controlled for. If one just used robust standard errors, the ten-percentage-point-higher NFRE share if there is a local business association might wrongly be treated as a precisely estimated effect, when, in fact, after allowing for the clustered sample design, one cannot rule out a zero effect.

The ways to correctly calculate variances from complex sample designs fall into two general categories: Taylor series linearization and replication methods. With the Taylor series approach, the variance estimate for a linear approximation to the estimator is used because many complex sample designs lack closed-form algebraic expressions for variances. These linearized variance estimators have been available for several decades, and are discussed, for example, in Kish and Frankel (1974). The replication methods take repeated sub-samples, or replicates, from the sample, either by bootstrapping or by jackknifing (leaving out observations, one-at-a-time, and recalculating the statistic) and use the standard deviation across the replicate estimates as an empirical estimate of the standard error. The replicates can be drawn by obeying the complex sample design, and may be especially useful when there are few (< 30) clusters (Cameron and Miller 2015). Notably, the variance estimators for dealing with complex sample designs have been available in *Stata* since the mid-1990s, and so there is little excuse for applied economists to continue to ignore these features of their samples when reporting regression estimates and conducting hypothesis tests.

There are two pernicious effects of ignoring complex sample designs. The first, shown above, is that some null effects will be wrongly treated as non-zero, and this may lead to ineffective interventions that are motivated by misleadingly precise results. Another effect of failing to report DEFFs is that under-powered sample designs are not exposed to scrutiny. Even with the budgetary logic of clustered samples, a given level of survey resources may be allocated by using statistically more efficient sampling plans; this scope for improvement would be more easily realized if DEFFs were reported and discussed. Other parts of agricultural economics, such as the choice experiments literature, pay attention to efficient sample designs (e.g., Ferrini and Scarpa 2007). Likewise, the impact evaluation literature has some consideration of how to best deploy survey resources (e.g., McKenzie 2012). Thus, there are examples to follow of improved understanding of sample designs and their impact on the estimated precision of empirical research findings.

Conclusions

The guidance to be gained from empirical research is only as strong as the weakest link in the chain connecting theory to research design to data to estimation, and to statistical hypothesis testing. For the last decade applied economists have focused on strengthening one link in this chain, choosing more credible research designs or identification strategies. Higher status has been accorded to research designs—like RCTs—that may plausibly identify causal effects. Yet absent attention to other links in the chain such as data,

survey) has DEFFs of 11.3 at national level, and DEFFs range from 9.5 to 13.7 at the regional level. I thank Kristen Himelein for this example.

the overall reliability of applied economic research may be no greater after the so-called “credibility revolution” than it was before. If applied economists followed their own textbook advice, they would direct their efforts to areas where marginal benefits most exceed marginal costs.

While it is possible that research design is still the link in the chain where marginal benefits of making further changes (such as replacing more observational studies with RCTs) most exceeds marginal costs, that case has hardly been demonstrated by the *randomistas*. If one concern is that researchers are estimating the wrong thing by using weak research designs—such as using OLS with observational data, and overstating returns to migration by about 30% (McKenzie et al. 2010)—then an obvious rejoinder is that there are lots of areas in applied economics where researchers seem to estimate the wrong thing. Raising the quality of research in some of these areas may be easier than switching research designs, especially considering the expense of RCTs and their ethical issues, although careful researchers would, ideally, make progress at the multiple points in the chain where our understanding may be constrained by weakness in theory, data, or research design.

In this paper I give five examples of areas where researchers in applied economics seem to estimate the wrong thing. Some have effects that are plausibly much larger than the effects of using randomized versus observational research designs. For example, mixing together quality responses to price with quantity responses seems to provide answers that are about 300% away from the right answers. Likewise, neglecting to allow for indirect spatial spillover effects may understate impacts of causal variables on outcomes by about two-thirds. While the errors associated with wrong choice of Tobit or using probits when one has continuous data may be smaller, these are also cheaper errors to correct, given that they require no new data, unlike for measuring quality responses to price or for measuring spillovers. When it is time to surround estimated impacts of covariates with confidence intervals, the tendency of applied economists to ignore the complex designs of the samples they use might lead to greatly exaggerated precision. An overly-narrow focus on identification strategies may lead to the neglect of these examples, and of other ways to improve the credibility and usefulness of applied economics research.

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